

TECHNOLOGY EVIDENCE BRIEF

TORR RF.

Toroidal multipolar radiofrequency — why a toroidal geometry delivers what conventional RF cannot.

EVIDENCE BASE

Patent No. 10-2113199 · 15+ peer-reviewed studies · clinical temperature data

BRITZMEDI CO., LTD.

CONFIDENTIAL

2026-04

▪ THE PROBLEM WITH CONVENTIONAL RF

Three architectures. Each structurally limited.

MONOPOLAR

Deep, but uncontrollable.

e.g. Thermage

Current travels from a single electrode through the body to a grounding pad. Energy passes through skin, fat, muscle and organs indiscriminately.

REPORTED RISKS

Grounding-pad burns · fat atrophy · significant pain. Complication rate ~10%.

BIPOLAR

Safe, but locked at the surface.

e.g. Venus Viva

Current flows only between two fixed electrodes on the handpiece. No grounding pad — but the electrodes are physically fixed.

STRUCTURAL LIMIT

Penetration depth locked at **1–4 mm**. Cannot reach subcutaneous fat. Superficial only.

NEEDLE RF

Deep — at the cost of invasion.

e.g. Morpheus8 · Potenza

Micro-needles penetrate the skin to deliver RF at depth. Effective, but invasive — bleeding, downtime, infection risk.

FDA SAFETY COMMUNICATION · OCT 2025

Burns · scarring · fat loss · facial disfigurement · nerve damage.

▪ THE PHYSICS — D/2 RULE

Why conventional bipolar depth is locked.

In bipolar RF, energy penetration depth equals approximately half the distance between electrodes. This is a well-established physical law.

THE IMPLICATION

To change depth, you must change *electrode spacing*. In conventional bipolar devices, that spacing is physically fixed — so depth cannot be adjusted. This is not a design flaw; it is a structural limitation of the two-electrode architecture.

D/2 RULE · BIPOLAR PENETRATION

Depth $\approx \frac{1}{2} \cdot d_{\text{(electrodes)}}$

Confirmed across six independent academic sources.

[1] PMC — 2020

[2] Medical Lasers — 2013

[3] IJDV — 2023

[4] Skin Research & Technology — 2023

[5] Health Science Reports — 2026

[6] ScienceDirect

THE TORR RF SOLUTION

Toroidal concentric electrodes — depth, electronically switched.

A center sphere surrounded by concentric ring electrodes — cross-section RF- | CH1 | CH2 | CH3 | CH2 | CH1 | RF-. The system electronically switches which electrode serves as the common (return) — changing electrode spacing without changing the operating principle. Always bipolar; no grounding pad. PATENT NO. 10-2113199.

EYE HANDPIECE

Φ 25 mm
center + 1 ring · 1 pair



FACE HANDPIECE

Φ 48 mm
center + 3 rings · 3 pairs



BODY HANDPIECE

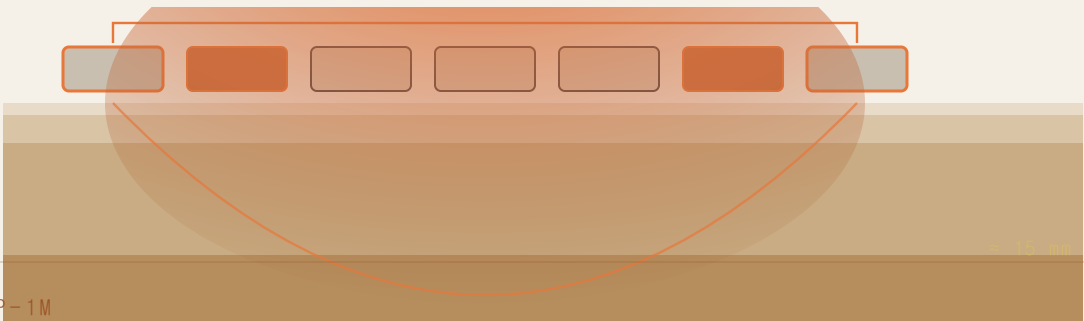
Φ 56 mm
3 pairs · auto-rotation MOTION



DEEP MODE · FARTHEST PAIRS

Outermost ring becomes common.

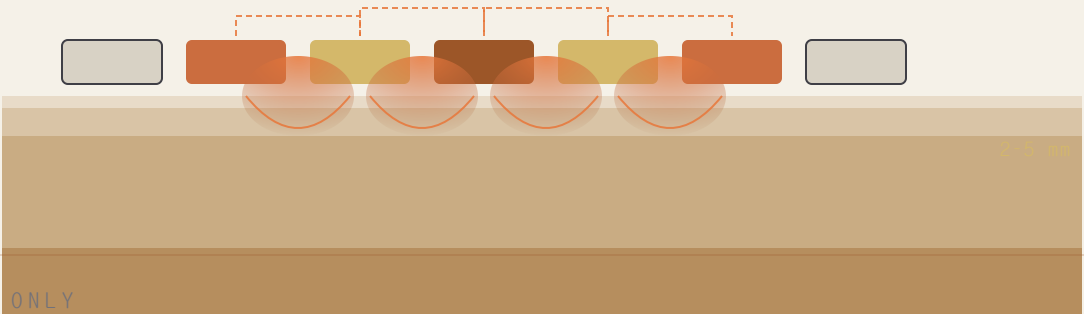
Maximum electrode spacing activates. The electric field forms a deep semicircular arch reaching the subcutaneous layer.



LOW MODE · ADJACENT PAIRS

Neighbours pair, +/- alternating.

Minimum spacing. Energy concentrates in the upper dermis as multiple shallow arcs.



▪ HOW TORR RF SOLVES EACH PROBLEM

Five problems. Resolved structurally.

01

MONOPOLAR BURN RISK

No grounding pad exists.

Purely bipolar. Current flows only between concentric electrodes on the handpiece. Pad burn is structurally impossible.

02

BIPOLAR FIXED DEPTH

Electrode spacing is electronic.

Common-electrode position switches in real time, changing spacing without changing principle. One handpiece covers superficial dermis to subcutaneous fat.

03

NEEDLE RF · FDA WARNING

Completely non-invasive.

No needles, no skin penetration. Zero bleeding, zero infection risk, zero downtime. None of the FDA's 2025 adverse events can structurally occur.

04

HOTSPOT BURNS

Distributed energy + NTC sensing.

Multiple electrode pairs distribute energy across a wide area. Real-time NTC sensors auto-cut RF on threshold. Patent No. 10-2018-0065876.

05

MULTIPLE DEVICES NEEDED

One platform · three handpieces · zero consumables.

Eye Φ25 mm · Face Φ48 mm · Body Φ56 mm cover the entire body from one platform.

NO TIPS · NO CARTRIDGES · NO REPLACEMENT PARTS

CLINICAL TEMPERATURE EVIDENCE

40–45 °C, sustained. Measured, not modelled.

SKIN TEMPERATURE TEST · JULY 2022

9 SUBJECTS

3 HANDPIECE TYPES

ARM · ABDOMEN · THIGH

FLIR THERMAL IMAGING

10-MIN AVERAGE

42.7

° C

15-MIN AVERAGE

43.1

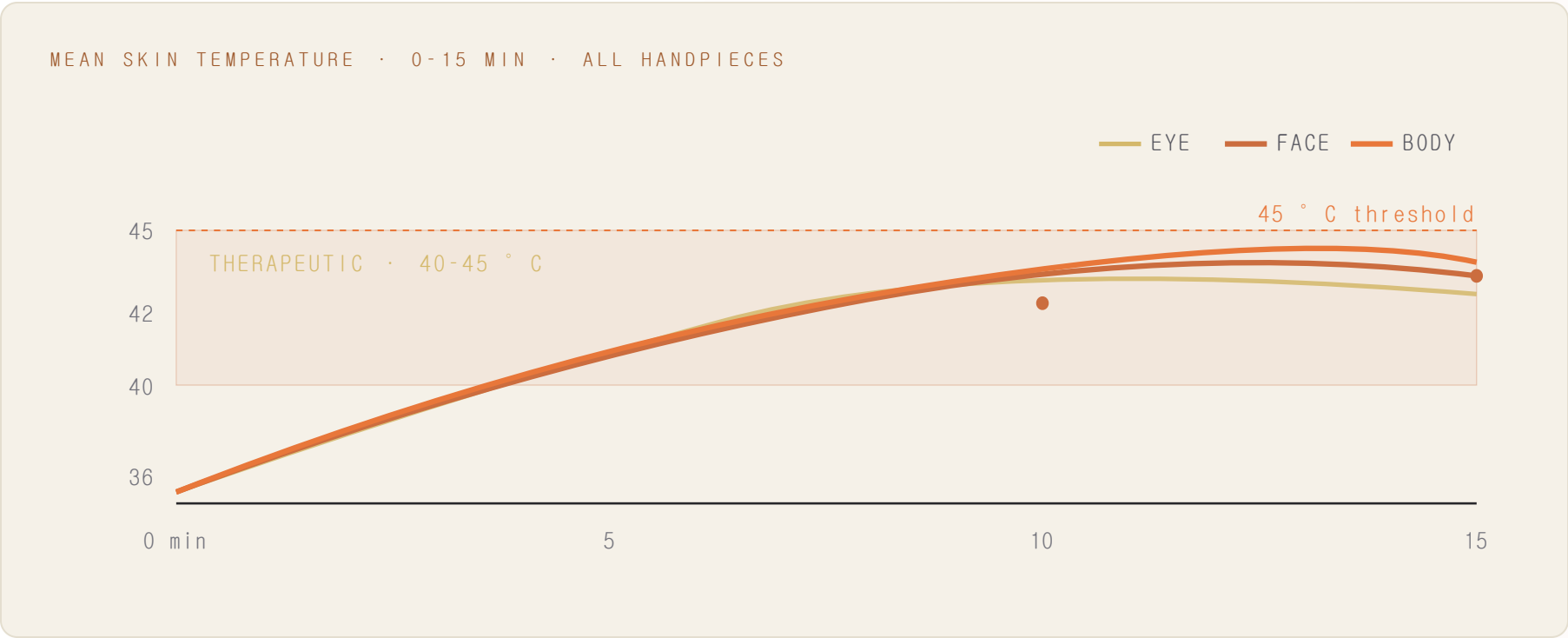
° C

ABOVE 45 ° C

0

INCIDENTS

All handpieces maintained the 40–45 °C therapeutic range across all sites. Zero overheating.



WHY 40-45 ° C MATTERS

Sustained therapeutic heat triggers neocollagenesis, elastogenesis and angiogenesis.

Peer-reviewed research confirms that maintaining skin temperature in the 40–45 °C window for ten minutes activates the collagen remodelling cascade.

9.0 % → 25.9 % ≈ 3 × collagen content
p = 0.018 · 2015 veterinary study

TORR RF maintains the same therapeutic window, measured in vivo across nine subjects.

■ CONCENTRIC MULTIPOLAR RF — PUBLISHED EVIDENCE

Three studies. One conclusion.

01

JCAD · 2019

JAPAN

Concentric-electrode multisource phase-controlled RF — patient experience.

PATIENTS

25

SATISFACTION

96 %

COMFORTABLE

100 %

Objective volume reduction confirmed by 3D imaging. Highest comfort rating across the cohort.

02

PMC · 2017

MULTICENTER

International multicenter trial — body contouring across 5–8 sessions.

IMPROVED

96 %

ABDOMINAL Δ

–2.9 cm

12-MO RETAINED

–1.9 cm

Confocal microscopy showed increased epidermal thickness and youthful collagen patterns at follow-up.

03

PMC · 2025

MECHANISM

Sequential dual-mode RF — enhanced remodelling vs. single-mode.

MODES COMPARED

2

COLLAGEN Δ

↑ enhanced

ELASTIN Δ

↑ enhanced

Validates TORR RF's dual-depth (Deep + Low) approach delivered from a single handpiece.

COMPETITIVE COMPARISON

Seven criteria. Side-by-side.

CRITERION	MONOPOLAR	BIPOLAR	NEEDLE RF	TORR RF
PENETRATION DEPTH	Fixed deep	Fixed shallow	Needle-dependent	Variable · electronic switching
GROUNDING PAD	Required	Not needed	Not needed	Not needed
INVASIVENESS	Non-invasive	Non-invasive	Invasive · needles	Non-invasive
PAIN LEVEL	High	Low-medium	Medium-high	Virtually painless
DOWNTIME	Moderate	Minimal	Redness · crusting	Zero
FDA SAFETY WARNING	Fat atrophy reported	None	Oct 2025 formal warning	None
CONSUMABLES	Tip replacement	Medium	Cartridge replacement	Zero

■ SUMMARY

Four dimensions of **superiority.**

01 · EFFICACY

Variable depth from one handpiece.

Superficial dermis to subcutaneous fat — selectable in real time. Built on the D/2 Rule and the toroidal geometry.

PATENT NO. 10-2113199

02 · SAFETY

Non-invasive · sensor-guarded.

Bipolar architecture eliminates pad burn. Real-time NTC temperature sensors prevent hotspots. Zero open FDA safety concerns.

PATENT NO. 10-2018-0065876

03 · USABILITY

Three handpieces · zero pads · zero needles.

Eye Φ25 mm reaches periorbital areas most RF cannot safely treat. Face Φ48 mm and Body Φ56 mm cover the rest.

EYE · FACE · BODY

04 · ECONOMY

Zero consumables · pure-margin sessions.

Every procedure is revenue, not cost. Accelerated ROI. Pricing flexibility while maintaining superior margins.

NO TIPS · NO CARTRIDGES · NO REPLACEMENTS

CORE MESSAGE

TORR RF is the only **toroidal multipolar RF** system that electronically controls penetration depth on the D/2 principle — delivering *monopolar-level depth* with *bipolar-level safety*, in a non-invasive, zero-consumable platform.